

Temidayo (Temi) Adeluwa

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PROFESSIONAL SUMMARY

Computational biologist with a pharmacology background and 5+ years of machine learning experience applied to genomics and clinical data to explain phenotypes. Driven by and dedicated to science, learning and teaching, mentoring, and volunteering, and looking to bring these into a pharma role where I can translate science into impact at scale, bring medications closer to patients, and contribute to organizational growth.

EDUCATION

Ph.D. Genetics, Genomics, and Systems Biology | The University of Chicago, IL Expected Nov 2026
[Dissertation]: *Explaining the Mechanisms of Disease-Associated Variants Using Deep Learning-Informed Predictors*

M.Sc. Biomedical Sciences | University of North Dakota, ND Jul 2021
[Thesis] *Using Machine Learning on Diverse Datasets to Predict Drug-Induced Liver Injury*

B.Sc. Pharmacology | College of Medicine, University of Lagos, Lagos, Nigeria Dec 2016
First class (Honors) | Rank: 1/~70 in department
[Final Year Project]: *Novel Action of Vinpocetine in the Prevention of Paraquat-Induced Parkinsonism in Mice: Involvement of Oxidative Stress and Neuroinflammation*

RELEVANT CLASS

Entrepreneurship in Healthcare and Life Sciences (BUSN 34210) | Booth School of Business Sep 2023 - Dec 2023

- Applied healthcare business fundamentals including drug development pathways, regulatory approval, market access, and biopharma funding models to assess clinical-stage risk and scientific strength of Cerevel Therapeutics' drug candidates (schizophrenia, Parkinson's disease, epilepsy pipeline) as part of a team analysis.
- The team's recommendation to approve the company and its products was validated when AbbVie acquired Cerevel for \$8.7B about 8 months later (Aug 2024).

LEADERSHIP & SERVICE

Co-facilitator, Growing Together: A Dinner Series | The University of Chicago Sep 2024 - Apr 2026

- Organized weekly well-being events for over 90 graduate students to foster community, friendship, and a sense of belonging to promote mental health.
- Led discussion sessions on well-being topics such as vulnerability, self-compassion, and finding joy.

Volunteer, Hyde Park Refugee Project | Hyde Park, Chicago, IL Jan 2024 - Present

- Tutored a 9th grade student of an Afghan Refugee family on a weekly basis in math, English, biology, theatre, U.S. history, chemistry, and other subjects.
- Successfully supported student in passing all classes up until grade 12.

Volunteer, Augustana Community Breakfast | Hyde Park, Chicago, IL Dec 2023 - Present

- Prepared, cooked, served, and cleaned up for 48+ (varies every month) individuals, offering a safe and welcoming space.

Volunteer, Nourishing Hope Chicago | Chicago, IL Nov 2023 - Present

- Packaged and distributed food and fresh produce for delivery to individuals in need.

Member of the Program in Computational Biology (PCB) | The University of Chicago Jul 2023 - Present

- Organized retreats and programming for the scientific community.
- Revamped the program's website, and coordinated weekly seminars, including regular communication with invited speakers.

- Discussion Leader**, Statistical Genetics | The University of Chicago Jan - Mar 2023
Jan - Mar 2024
- Led discussion sessions and tutorials for ~20 students across an undergraduate section (Jan–Mar 2023) and a graduate section (Jan–Mar 2024), covering statistical frameworks, R, Python, Bash, and HPC cluster usage.
- Hackathon Instructor** | Program in Computational Biology, The University of Chicago Apr 2022, Mar 2023
- Developed and led hands-on lesson plans for Hackathon 2022 (gene expression prediction with Enformer) and Hackathon 2023 (building a minimal GPT in PyTorch) for over 50 professors and graduate students.
- Assistant Governor and Peer Instructor on Financial Inclusion Matters**, Community Development Service (CDS) Group | Lagos, Nigeria Jun 2017 - Apr 2018
- Taught secondary school students financial literacy and ensured compliance and accountability of CDS group members.
- Director of Studies**, Department of Pharmacology | University of Lagos, Lagos, Nigeria Jan 2016 - Dec 2016
- Organized and taught pharmacology and chemistry classes to 100+ students in pharmacy, physiology, and pharmacology departments.

RESEARCH EXPERIENCE

- Graduate Research Assistant** | The University of Chicago, Chicago, IL Sep 2021 - Present
Advisor: Dr. Hae Kyung Im
- Developed Enpact, a machine learning method to predict transcription factor (TF) binding using sequence-based epigenomic information and genotype data. Enpact predicts TF binding for 200+ TFs across 54 tissues with AUPRC ranging from 0.68 to 0.99.
 - Identified 164 significant androgen-receptor binding sites (ARBS), 2.2 times more than CWAS (74 ARBS), including 118 loci uniquely detected by Enpact-based association testing.
 - Integrated Enpact with genome-wide association statistics into TFXcan, a scalable tool for testing associations between TF binding and phenotypes in different biological contexts, even in datasets with small sample sizes. *Manuscript in press.*
 - Identified transcription factor drivers of diseases such as prostate cancer, childhood-onset asthma, and type 2 diabetes.
 - Released TFXcan as an open-source framework (github.com/hakyimlab/TFXcan) for researchers to apply to GWAS datasets.
- Graduate Research Assistant** | University of North Dakota, Grand Forks, ND Aug 2019 - Jul 2021
Advisor: Dr. Junguk Hur
- Applied machine learning to predict postpartum depression (PPD) in maternal women, identifying key clinical and sociodemographic risk factors, including breastfeeding duration, prior depression, and maternal age.
 - Co-developed a novel algorithm combining Particle Swarm Optimization and Gradient Boosting Machines for cell segmentation, achieving improved sensitivity (0.90) and accuracy (0.849) over baseline methods (0.831 and 0.807, respectively); presented at the 2021 IEEE SSCI.
 - Integrated gene expression, drug exposure, and adverse event data to build ensemble machine learning models for predicting drug-induced liver injury, which formed the foundation of my master's thesis and was presented at international conferences.
- Undergraduate Researcher** | College of Medicine, University of Lagos, Lagos, Nigeria Jul 2014 - Oct 2016
Co-advisors: Dr. Ismaila Ishola & Dr. Akinwunmi Akinyede
- Final year capstone project: Investigated vinpocetine's mechanisms in a mouse model of Parkinsonism, contributing to a peer-reviewed publication.

ADDITIONAL WORK EXPERIENCE

Graduate Trainee Sterling Bank Plc., Lagos, Nigeria	Feb 2019 - Jul 2019
Content Writer and Head of Admissions of Social Media & Email Marketing Hotels.ng	Aug 2018 - Jan 2019
Hospital Pharmacist 68 Nigerian Army Reference Hospital, Yaba, Lagos, Nigeria	Jul 2017 - Apr 2018
Locum/Community Pharmacist Luxe Pharmacy, Sabo Yaba, Lagos, Nigeria	Jul 2017 - Aug 2017

AWARDS

• UChicago GRAD Computational Biology Trek	Aug 2025
• Bruce Weir Summer Institute in Statistical Genetics Travel Scholarship	May 2025
• Google CS Research Mentorship Program	Jun 2021
• University of Lagos Outstanding Scholars Award (\$150 annually for four years)	2012 - 2016
• Provost Award for the Best Graduating Student in Pharmacology, 2016/2017 Academic Session	2016
• Certificate of Excellence/Merit for the Best Graduating Students of the College of Medicine and the Faculty of Pharmacy	2016
• Association of Pharmacology Students of the University of Lagos Award for the Most Outstanding Student	2015

PRESENTATIONS

- *TFXcan reveals transcriptional programs driving complex traits and disease.* The Biology of Genomes, Long Island, NY. May 2025. (Poster)
- *Regulatory Insights into Type 2 Diabetes using Genetic Predictors of Transcription Factor Binding.* Accelerating Medicines Partnership® Program for Common Metabolic Diseases (AMP CMD) Parliament Meeting, Philadelphia, PA. 2025. (Invited speaker)
- *Personalized transcription factor binding from deep learning algorithms offers a new framework to identify disease associations.* The Biology of Genomes, Long Island, NY. May 2023. (Poster)
- *Personalized transcription factor binding from deep learning algorithms sheds light into the cis and trans regulation of gene expression.* American Society of Human Genetics. Nov 1–5, 2023. (Poster)
- *GBMPSO: Hybrid Gradient Boosting Machines with Particle Swarm Optimization in Cell Segmentation Data.* 2021 IEEE Symposium Series on Computational Intelligence (SSCI), Orlando, FL. 2021. (Invited talk)
- *Predicting Drug-Induced Liver Injury with Machine Learning Incorporating a Diverse Set of Public Datasets.* Cold Spring Harbor Laboratory (CSHL) Biological Data Science, Long Island, NY. Nov 4–6, 2020. (Poster)
- *Predicting Drug-Induced Liver Injury with Machine Learning Incorporating a Diverse Set of Public Datasets.* Annual International Conference on Critical Assessment of Massive Data Analysis (CAMDA), Virtual. Jul 13–14, 2020. (Oral)
- *Standard Prescription Writing (A Call for Teamwork).* 68 Nigerian Army Reference Hospital, Yaba, Lagos, Nigeria. Oct 20, 2017. (Oral, presented to hospital command staff)
- *Recent Developments in Adenosine Receptor Ligands and Their Potentials as Novel Drugs.* Department of Pharmacology, University of Lagos. Aug 25, 2016. (Seminar)
- *Peptide Therapeutics: Current Status and Future Directions.* Department of Pharmacology, University of Lagos. Jul 22, 2016. (Seminar)

TECHNICAL SKILLS

[Programming languages and coding tools] R, Python (pandas, matplotlib), Bash/Shell, SQL, Jupyter/Jupyter Lab, Quarto

[Bioinformatics] PLINK, samtools, GATK, BioPython, bcftools, and genomic data analysis software

[Machine learning] Caret, glmnet, mlr, scikit-learn, NumPy, PyTorch, Transformers/LLM experience

[Computing] High-performance computing (HPC), parallel programming techniques, Google Cloud Computing

[Pipeline development] Snakemake

[Containerization, reproducibility, and Version Control] Singularity/Apptainer, Docker, Git/GitHub

[AI-assisted tools] LLM-based coding assistants (Claude), prompt engineering

CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT

- Data Science: Foundations using R (Coursera; Johns Hopkins University)
- Artificial Intelligence Fundamentals (Data Science & Artificial Intelligence Academy; Sterling & Coven Works)
- Professional Certificate in Data Science (edX; Harvard University)
- Applied Data Science with Python Specialization (Coursera; University of Michigan)
- Mathematics for Machine Learning (Coursera; Imperial College London)
- Learn SQL Basics for Data Science (Coursera; UC Davis)
- Python Programmer (DataCamp)
- Google IT Automation with Python (Coursera; Google)
- Bioinformatics Methods I & II (Coursera; University of Toronto)
- Deep Learning Specialization (Coursera; deeplearning.ai)
- TensorFlow 2 for Deep Learning Specialization (Coursera; Imperial College London)
- Object-Oriented Programming in Java Specialization (Coursera; UC San Diego, Duke University)

PUBLICATIONS

- Adeluwa T, et al. TFXcan reveals transcription factor programs driving GWAS loci (*Manuscript in press*).
- Zhou Y, Adeluwa T, et al. scPrediXcan: A cell-type-specific transcriptome-wide association study framework. *Cell Genomics*. 2025.
- Adeluwa T, Kim E, Hur J. GBMPSO: Hybrid Gradient Boosting with Particle Swarm Optimization for Cell Segmentation. 2021 IEEE SSCI.
- Adeluwa T, McGregor BA, Guo K, Hur J. Predicting Drug-Induced Liver Injury Using Machine Learning. *Front Pharmacol*. 2021.
- Shin D, Lee KJ, Adeluwa T, Hur J. Machine Learning-Based Modeling of Postpartum Depression. *J Clin Med*. 2020.
- Ishola IO, Akinyede AA, Adeluwa T, Micah C. Novel action of vinpocetine in the prevention of paraquat-induced parkinsonism in mice: involvement of oxidative stress and neuroinflammation. *Metab Brain Dis*. 2018